TAPO: Turn-Adaptive Policy Optimization for Multi-Turn Language Model Reinforcement Learning

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Abstract

Multi-turn reinforcement learning is increasingly important as language agents use longer contexts, tools, search, and interactive environments. With only final-reward supervision, PPO and GRPO form clipped token losses from reward-derived advantages and aggregate them across response tokens. A long turn can therefore receive disproportionate update mass even when its tokens repeat one underlying decision, amplifying turn-level noise rather than independent evidence. We propose TAPO (Turn-Adaptive Policy Optimization), a turn-level-credit-free algorithm that estimates the relative turn- and token-level variance of the clipped token losses and adaptively interpolates between token-normalized and turn-normalized aggregation. Across four search benchmarks (including OOD Bamboogle) and two embodied-interaction benchmarks under both GRPO- and PPO-style training, TAPO improves the mean of all 72 reported pass@k and avg@k values ($k \in \{1, 2, 4\}$) over the corresponding token-normalized (Token) baseline by 3.5 points for the 9B model and 4.0 points for the 4B model. The gains are 4.9 and 5.1 points, respectively, on the search benchmarks, while pass@1 stays within 0.6 points of Token for 9B and 2.0 points for 4B on low-variance WebShop and ALFWorld.

1 Introduction

Reinforcement learning from preference or task rewards has become a central tool for improving language models after supervised training. PPO-style clipped policy-gradient objectives (Williams 1992; Schulman et al. 2017; Ouyang et al. 2022) and value-free variants such as GRPO (Shao et al. 2024) are especially attractive when cheap final rewards are available. (We use *clipped policy-gradient methods* to refer to this PPO/GRPO family.) This recipe was originally most natural for single-turn outputs, but language-model RL is increasingly multi-turn: longer context windows, tool calls, search, web navigation, and agentic workflows (Nakano et al. 2021; Schick et al. 2023; Yao et al. 2023) all create trajectories with many intermediate decisions before a final answer.

The central difficulty is credit assignment. Applying final-reward-derived advantages across trajectory tokens avoids explicit turn-level rewards, but in multi-turn trajectories it can be high variance. Token normalization treats each token as one unit of update mass, so a long turn receives a larger total weight simply because it contains more tokens, even though many tokens within a turn describe the same action. Existing turn-level credit methods reduce this ambiguity through sequential updates, dense intrinsic rewards, or teacher signals, but add computation, task-specific supervision, or proxy objectives; we review them in the next section. When the final outcome reward is the only trusted supervision, reducing variance without recovering explicit turn credit is therefore a useful research direction.

Our starting point is that a token loss in multi-turn RL contains both turn-level variation and token-level variation. A turn often corresponds to a coherent action: asking a question, issuing a tool call, observing a webpage, or writing a reasoning step. Tokens within that turn share the same latent decision and reward consequence. When turn-level variation dominates, token-normalized clipped objectives can overweight long turns and have high variance or collapse late in training. Pure turn normalization fixes the length bias, but it can be worse when independent token noise dominates. This paper asks a narrower question: using only the same clipped token losses and the same final reward, can we choose a lower-variance mix of token-normalized and turn-normalized aggregation?

We answer this question with TAPO (Turn-Adaptive Policy Optimization). TAPO first computes the same clipped token losses as the underlying backbone, then constructs token-normalized and turn-normalized aggregates. A soft variance-ratio gate interpolates between them, assigning more weight to turn normalization when shared turn-level structure dominates and returning toward the token-normalized objective when token-level noise dominates. We instantiate this mechanism with both GRPO- and PPO-style objectives and use the standard token-normalized version of each objective (the "Token" rows in Tables 2–5) as its direct baseline. Averaged over the six benchmarks, both metrics, both objectives, and $k \in \{1, 2, 4\}$, TAPO improves over Token by 3.5 points for 9B and 4.0 points for 4B. The search-only gains are 4.9 and 5.1 points; under GRPO/PPO they are 5.2/4.5 points for 9B and 3.8/6.3 points for 4B. On Bamboogle, the aggregate is 39.3 for TAPO versus 30.1 for Token at 9B, and 30.8 versus 24.0 at 4B. By contrast, pass@1 differs from Token by at most 0.6 points for 9B and 2.0 points for 4B on Web-

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Shop (Yao et al. 2022) and ALFWorld (Shridhar et al. 2021), whose measured turn-level variance ratios are low.

Contributions Our main contributions are summarized as follows:

- **Theoretical Analysis of Token- and Turn-Level Variance.** We provide a variance analysis of token-normalized clipped policy-gradient objectives in multi-turn RL by decomposing token-loss variance into shared turn-level noise and independent token-level noise. The analysis reveals a fundamental trade-off: token-normalized aggregation is efficient when token noise dominates, but it can overweight long turns and become high variance when turn-level correlation dominates.
- **Novel Algorithm with Turn-Adaptive Aggregation.** We propose TAPO (Turn-Adaptive Policy Optimization), an aggregation rule that combines token-normalized and turn-normalized clipped policy losses using a variance-ratio gate. TAPO keeps the same final-reward supervision as the backbone objective and does not require an additional critic, teacher reward model, additional actor update schedule, or turn-level reward labels, while also improving stability on out-of-distribution search evaluation when traditional clipped policy-gradient methods weaken late in training.
- **Evaluation Across Search and Embodied Multi-Turn Tasks.** We evaluate TAPO with both GRPO and PPO on 4B and 9B models across four search benchmarks and two embodied tasks. Relative to the token-normalized objective in each algorithm family, TAPO delivers consistent aggregate gains on the high-variance search evaluations and remains close to the backbone objective on WebShop and ALFWorld in the low turn-level variance-ratio regime. The Bamboogle checkpoint curve further shows that TAPO avoids the late-training collapse of the baseline clipped policy-gradient method.

2 Related Work

Allocating Turn-Level Reward Through Extra Compute One line of multi-turn RL work tries to recover more precise turn-level learning signals by changing the training procedure or adding dense intrinsic rewards. SeeUPO treats a multi-turn trajectory as a sequence of virtually separated bandit agents and performs reverse-order sequential actor updates, so later turns are optimized before earlier turns (Hu et al. 2026). This gives a clean backward-induction view and convergence guarantee, but the cost is several actor updates over one trajectory rather than one standard policy update. IGPO takes a different route: in search-style QA, it defines a turn reward by the marginal increase in the policy's probability of the ground-truth answer (Wang et al. 2025). This improves credit assignment, but it needs additional likelihood evaluations. Both methods introduce extra computation.

Teacher and Rubric-Based Turn Rewards A second line of work assigns turn-level reward with an external teacher, process reward model (Lightman et al. 2023), or hand-designed rubric. For example, TIPS shapes each reasoning and tool-call segment by how much it increases a teacher model's likelihood of the correct answer (Xie et al. 2026). More generally, rubric-based approaches can ask a judge model (Zheng et al. 2023) whether a turn is helpful, grounded, or follows a desired tool-use pattern. These signals can be useful when the teacher is reliable, but they add inference cost and introduce a proxy objective. In long-horizon RL, the policy may learn to satisfy the teacher or rubric without improving the final environment reward, especially when the proxy rewards are easier to game than the task itself. Table 1 summarizes these trade-offs.

3 Preliminaries

Problem Formulation We consider an outcome-only multi-turn RL setting. A prompt or environment instance x produces a trajectory

$$\tau = (x, a_1, o_1, a_2, o_2, \ldots, a_K, o_K, r), \tag{1}$$

where K is the number of turns, a_k is the language-model reasoning/tool-use segment or environment action produced after receiving the previous observation, o_k is the tool or environment observation after that turn, and r is the final scalar reward.

Token-Normalized Clipped Policy Loss When the final reward is the only supervision that is trusted, exact causal credit for individual turns is expensive to recover. Clipped policy-gradient methods compute reward-derived token advantages and use one clipped policy-gradient update, without an extra teacher reward, rubric reward, or turn-level credit allocator. GRPO broadcasts a group-relative trajectory advantage, whereas PPO uses its return- and value-based advantage estimator.

Let t index valid response tokens in the trajectory. With old policy $\pi_{\theta_{\text{old}}}$ and current policy π_θ, define the token ratio

$$\rho_t(\theta) = \exp\left(\log \pi_\theta(y_t \mid y_{<t}, x) - \log \pi_{\theta_{\text{old}}}(y_t \mid y_{<t}, x)\right). \tag{2}$$

Let A_t denote the token advantage supplied by the backbone estimator; it is constant within a GRPO trajectory but may vary across tokens under PPO. The usual clipped policy-gradient token surrogate is

$$\ell_t = \max\left(-A_t\rho_t(\theta), -A_t \,\text{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\right), \tag{3}$$

optionally with dual clipping for negative advantages. Let N be the total number of valid response tokens in the trajectory. The baseline trajectory loss is the token mean

$$L^{\text{tok}} = \frac{1}{N}\sum_{t=1}^{N} \ell_t. \tag{4}$$

Thus, if a trajectory has turns T_k of lengths m_k, the token-normalized objective gives turn k total aggregation mass m_k/N. This length weighting is the source of the variance problem studied in this paper.

The weakness is that token count becomes the implicit credit-allocation rule. If a long turn contains many highly correlated tokens, token-normalized aggregation gives that turn more update mass even though those tokens may not provide proportional independent evidence. This can produce high-variance updates, late-training collapse, or overfitting to verbose turns when turn-level variation dominates token-level noise.

Table 1: Positioning of multi-turn RL credit-assignment strategies. TAPO keeps the outcome-only property of clipped policy-gradient baselines, while reducing turn-level variance by changing the trajectory aggregation rule when the batch statistics indicate that token-length weighting is high variance.

Method family	Turn-level signal	Extra cost	Outcome-only	Reduces turn-level variance	Main weakness
Clipped policy-gradient	Apply reward-derived advantages to tokens	**No**	**Yes**	**No**	Length-weighted turns can be high variance
SeeUPO	Sequential turn update through backward induction	**Yes**	**Yes**	**Yes**	Several actor updates are costly
IGPO	Information-gain reward per turn	**Yes**	**Yes**	**Yes**	Extra forward compute
Teacher/rubric methods	Judge or teacher reward per turn	**Yes**	**No**	**Yes**	Expensive proxy can be hacked
TAPO	**Adaptive aggregation**	**No**	**Yes**	**Yes**	Requires turn boundaries

Turn-Normalized Clipped Policy Loss One direct way to reduce the variance above is to use turn-normalized aggregation. In multi-turn tasks, response tokens are partitioned into K turns. Let T_k be the token set for turn k, with length $m_k = |T_k|$ and total length $N = \sum_{k=1}^{K} m_k$. Turn-normalized aggregation first averages token losses inside each turn and then gives equal mass to each turn:

$$L^{\text{turn}} = \frac{1}{K} \sum_{k=1}^{K} \frac{1}{m_k} \sum_{t \in T_k} \ell_t. \tag{5}$$

Equations (4) and (5) differ only in aggregation. The token-normalized objective assigns turn k weight m_k/N, while turn normalization assigns weight $1/K$. Turn normalization removes the length bias, but it is not always better: when independent token-level noise is large, averaging over all tokens can have lower variance, and giving short noisy turns equal mass can increase variance.

4 Method

4.1 Variance Model

The purpose of the variance model is to decide, separately for each trajectory, whether to use token-normalized or turn-normalized loss aggregation. Consider one trajectory with K turns. The subscript $k \in \{1, \dots, K\}$ indexes the turn, and the subscript t indexes a token inside turn k. Let m_k be the number of valid tokens in turn k, and let $N = \sum_{k=1}^{K} m_k$ be the total number of valid response tokens in the trajectory. Let $x_{k,t}$ denote a scalar token-loss contribution. We model

$$x_{k,t} = \mu + u_k + \epsilon_{k,t}, \tag{6}$$

where μ is the shared trajectory-level signal, u_k is the turn-level deviation, and $\epsilon_{k,t}$ is token-level residual noise. We write $\tau^2 = \text{Var}(u_k)$ for the turn-level variance and $\sigma^2 = \text{Var}(\epsilon_{k,t})$ for the token-level variance; both are estimated from that trajectory's detached token losses. The shared signal μ receives total weight one under both aggregation rules, so it does not affect the variance comparison.

Let $\bar{x}_k = m_k^{-1} \sum_{t \in T_k} x_{k,t}$ denote the average token contribution in turn k, where T_k is the token set of that turn. This is just a shorthand for the per-turn mean that turn-normalized aggregation operates on. Then

$$\bar{x}_k = \mu + u_k + \bar{\epsilon}_k, \qquad \text{Var}(\bar{x}_k) = \tau^2 + \frac{\sigma^2}{m_k}. \tag{7}$$

In practice, turn boundaries are already known from the rollout phase, such as one search turn in a search task or one action turn in an embodied task, and the backbone has already computed the clipped token losses from its own advantage estimator. We can therefore compute τ^2 and σ^2 without extra model passes; the full estimator details are given in Appendix B. The term σ^2/m_k is the token-noise variance that remains in the mean of a turn with m_k tokens. Thus the empirical variance of turn means contains both turn-level variation and this residual token noise; the estimator for τ^2 subtracts the estimated token-noise contribution so that short turns are not mistaken for high turn-level variance. Token-normalized aggregation averages over all tokens, so turn k receives weight m_k/N. Turn-normalized aggregation first averages within each turn and then averages turns, so every turn receives weight $1/K$. The resulting variance comparison is summarized below; the complete proof is in Appendix B.

Proposition 1 Assume the unified model in Eq. (6), zero-mean independent turn deviations, independent token residuals, and fixed turn lengths $m_1, \dots, m_K$. The token-normalized and turn-normalized trajectory losses are

$$L_{\text{tok}} = \sum_{k=1}^{K} \frac{m_k}{N} \bar{x}_k, \qquad L_{\text{turn}} = \sum_{k=1}^{K} \frac{1}{K} \bar{x}_k. \tag{8}$$

Their variances are

$$\text{Var}(L_{\text{tok}}) = \tau^2 \sum_{k=1}^{K} \left(\frac{m_k}{N} \right)^2 + \frac{\sigma^2}{N}, \tag{9}$$

$$\text{Var}(L_{\text{turn}}) = \frac{\tau^2}{K} + \frac{\sigma^2}{K^2} \sum_{k=1}^{K} \frac{1}{m_k}. \tag{10}$$

The first term in each equation is caused by turn-level deviation; the second term is caused by token-level residual noise.

The proposition shows that the two estimators spend variance in different places. Token-normalized aggregation gives more mass to long turns, which is statistically useful only

when long turns reduce independent token noise. Turn-normalized aggregation ignores length, which is statistically useful only when the turn is the effective independent unit. Turn-normalized aggregation is predicted to have lower variance when $\mathrm{Var}(L_{\mathrm{tok}}) > \mathrm{Var}(L_{\mathrm{turn}})$, equivalently

$$\tau^2 \left(\sum_{k=1}^{K} \left(\frac{m_k}{N} \right)^2 - \frac{1}{K} \right) > \sigma^2 \left(\frac{1}{K^2} \sum_{k=1}^{K} \frac{1}{m_k} - \frac{1}{N} \right). \tag{11}$$

This inequality yields the trajectory-level variance ratio

$$R = \frac{\tau^2 \left[\sum_{k=1}^{K} \left(\frac{m_k}{N} \right)^2 - \frac{1}{K} \right]}{\sigma^2 \left[\frac{1}{K^2} \sum_{k=1}^{K} \frac{1}{m_k} - \frac{1}{N} \right] + \varepsilon}. \tag{12}$$

Equation (11) and the ratio in Eq. (12) are the central diagnostic. The numerator is the turn-level variance reduction from replacing length weights m_k/N with equal turn weights $1/K$. It is large when turn lengths are imbalanced and τ^2 is large. The denominator is the token-noise cost from the length term $\frac{1}{K^2} \sum_{k=1}^{K} \frac{1}{m_k}$, minus the token-normalized baseline $\frac{1}{N}$. It is large when turns are short or uneven and σ^2 is large. Thus the model predicts exactly the behavior we want: use more turn normalization in high turn-variance regimes, and fall back toward token-normalized clipped policy losses when token noise dominates.

4.2 TAPO: Turn-Adaptive Policy Optimization

The condition in Eq. (11) suggests a cheap trajectory-level gate for choosing the aggregation method. TAPO computes one detached gate for each trajectory, using only that trajectory's valid token losses and turn lengths, and uses it to interpolate between the two aggregates. The minibatch objective is the mean of these trajectory-level losses.

Soft Gate We compute the ratio between the estimated benefit of turn normalization and the estimated cost of token noise:

$$R = \frac{\hat{\tau}^2 B}{\hat{\sigma}^2 C + \varepsilon}, \qquad \lambda = \frac{R}{1+R},$$
$$\text{where } B = \sum_{k=1}^{K} \left(\frac{m_k}{N} \right)^2 - \frac{1}{K}, \quad C = \frac{1}{K^2} \sum_{k=1}^{K} \frac{1}{m_k} - \frac{1}{N}. \tag{13}$$

Here the soft gate λ is produced from the estimated variance ratio R in Eq. (12). We use a soft gate rather than a hard switch so that the objective changes smoothly as the trajectory statistics change. The gate is the interpolation weight: $\lambda = 0$ means pure token-normalized aggregation, $\lambda = 1$ means pure turn-normalized aggregation, and intermediate values mix them smoothly. Degenerate-trajectory handling is specified in Appendix A.

Training Objective Let L_{tok} denote the token-normalized clipped policy-gradient aggregate in Eq. (4), and let L_{turn} denote the turn-normalized aggregate in Eq. (5). Using the soft gate, we define the final loss as

$$L_{\mathrm{TAPO}} = (1-\lambda) L_{\mathrm{tok}} + \lambda L_{\mathrm{turn}}. \tag{14}$$

Intuition From Special Cases Figure 1 visualizes the Token and Turn endpoints and the TAPO interpolation between them.

If all turns have equal length, then $m_k/N = 1/K$ and the turn-level term in Eq. (11) vanishes. In that case, the two aggregation rules have nearly the same turn weighting, and token-normalized aggregation is preferred because it does not pay the extra token-noise cost of giving short turns special treatment. This explains why the TAPO gate should stay close to the token-normalized backbone when the trajectory has balanced turns or weak turn structure.

If one turn is much longer than the others, token-normalized aggregation can assign that turn far more mass than $1/K$. This is harmless when the long turn is genuinely more reliable because it averages many independent token residuals. It is harmful when the long turn is only verbose: the additional tokens repeat the same latent decision, so they carry shared turn-level noise rather than independent evidence. In this regime, the length imbalance factor $\sum_k (m_k/N)^2 - 1/K$ becomes large, and turn-normalized aggregation can reduce variance by preventing the long turn from dominating the trajectory update.

The opposite regime is also important. A very short turn has a noisy mean because $\mathrm{Var}(\bar{x}_k)$ contains the residual term σ^2/m_k. Pure turn normalization gives such turns equal mass even when their token averages are unreliable. This is why the method does not simply replace token-normalized training with turn-normalized training. The gate compares the estimated turn-level benefit to the token-noise cost, so the algorithm can interpolate rather than choosing one estimator globally.

5 Experiments

5.1 Experimental Settings

Benchmarks and Evaluations We evaluate TAPO on two categories of multi-turn tasks and jointly train each policy on a mixture spanning both categories. The training mixture contains 2WikiMultiHopQA (Ho et al. 2020), MuSiQue (Trivedi et al. 2022), and HotpotQA (Yang et al. 2018), which use the Wikipedia Python package as the search tool (Goldsmith 2014); WebShop (Yao et al. 2022), where the model searches for and purchases products that satisfy an instruction; and ALFWorld (Shridhar et al. 2021), where the model completes embodied tasks through text actions. We evaluate the final jointly trained checkpoint separately on each benchmark's own validation set. Bamboogle (Press et al. 2023) is excluded from the training mixture and used as a compositional OOD evaluation benchmark.

This evaluation separates high-variance search trajectories from embodied interaction despite their joint training. Search requires repeated decisions about reasoning, tool calls, and evidence integration; Bamboogle additionally tests whether those decisions generalize beyond the training mixture. WebShop and ALFWorld instead require repeated state-conditioned actions and exhibit much lower measured turn-level variance ratios. Tables 2 and 3 report GRPO-family results, while Tables 4 and 5 report PPO-family results. For $k \in \{1, 2, 4\}$, pass@k is the fraction of examples with at

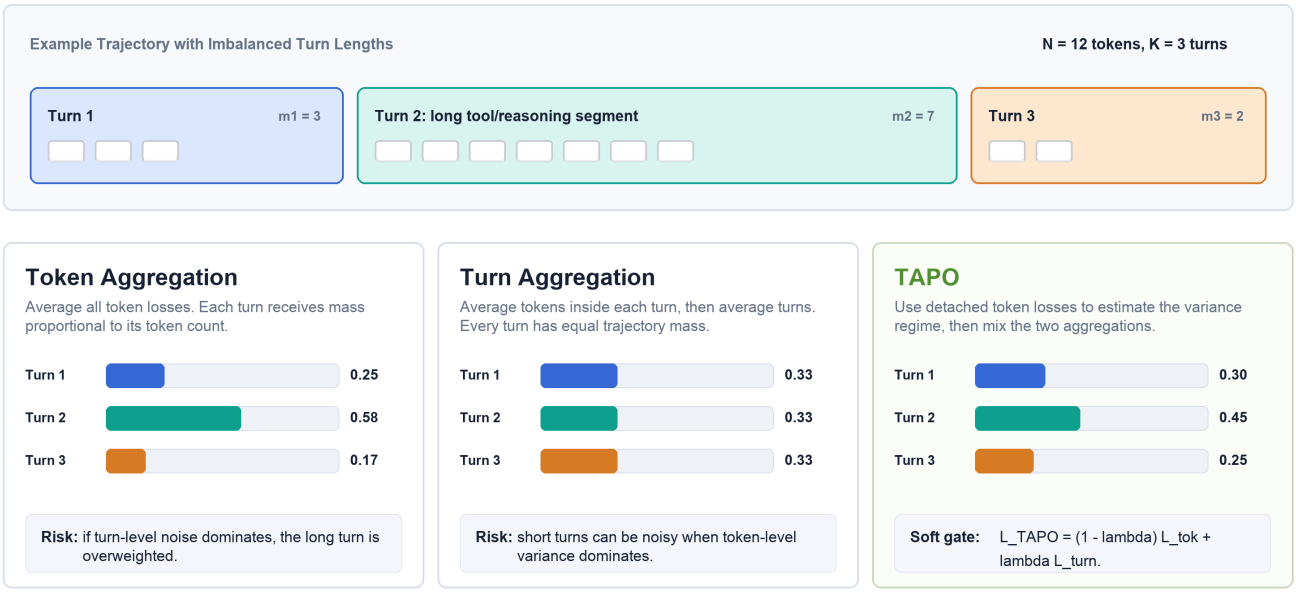


Figure 1: Token aggregation, turn aggregation, and TAPO on a toy trajectory with imbalanced turn lengths. Token aggregation weights turns by token count, turn aggregation gives each turn equal mass, and TAPO mixes the two with a detached variance gate.

least one successful rollout among k samples, and avg@k is the mean task reward across those rollouts. WebShop and ALFWorld use task success as the final reward.

Baselines and Backbone Models The reported comparisons use Qwen3.5 4B and 9B backbone models (Qwen Team 2026). Within each algorithm family, *Token* is the primary baseline: GRPO or PPO-style training with token-normalized loss aggregation. The GRPO-style runs use group-relative trajectory advantages, while the PPO-style runs use PPO's return- and value-based advantage estimator. *Turn* changes only the aggregation to the always-on turn-normalized objective ($\lambda = 1$), while TAPO adaptively mixes the Token and Turn objectives. The GRPO family additionally includes GIGPO (Feng et al. 2025) as a representative multi-turn turn-credit baseline; Section 6.2 explains this choice and reports the direct comparison. Thus, all point improvements reported below compare TAPO with the Token row of the same model scale and algorithm family unless another baseline is named explicitly.

Implementation Details The GRPO clipping parameter ϵ is set to 0.2 for both lower and upper clipping bounds. The KL penalty coefficient is 0.001. We sample 8 rollouts per instance during training. For the search setting, we randomly choose 3,000 training samples from each source: HotpotQA, MuSiQue, and 2WikiMultiHopQA. The validation set size is 500, and the learning rate is 1×10^{-6}. The numerical constant ε in the variance ratio R is set to 10^{-6}. For WebShop, we choose 5,000 items as environments and generate 1,657 training examples; the validation set size is 500. For ALFWorld, we choose 2,000 games and use a validation set of 500.

We train all models for three epochs to ensure full convergence. Tables 2–5 report validation results from the final checkpoint after training. For the search tasks, one turn is defined as one thinking segment plus one tool call. For embodied tasks such as WebShop and ALFWorld, one turn is defined as one thinking segment plus one chosen action.

Infrastructure All experiments are conducted on clusters of NVIDIA H200 GPUs with 144GB memory. The implementation is built on PyTorch (Paszke et al. 2019) and uses the veRL library for distributed RL training infrastructure (Sheng et al. 2024). The agent-related infrastructure is built on OpenAI Gym-style environments (Brockman et al. 2016).

5.2 Main Results

Final Results Tables 2–5 compare TAPO with the Token baseline at both model scales and under both policy-optimization families. For a compact summary, we average the pointwise TAPO–Token differences across all six tasks, GRPO and PPO, pass@k and avg@k, and $k \in \{1, 2, 4\}$. This gives an average improvement of 3.5 points for 9B and 4.0 points for 4B. Because each aggregate covers 72 reported values, it summarizes performance across algorithms, sampling budgets, and task types rather than selecting a single favorable metric.

The gains concentrate on search. Averaged over the four search benchmarks and both algorithm families, TAPO improves by 4.9 points for 9B and 5.1 points for 4B. Broken down by objective, the search gains are 5.2 and 4.5 points for 9B under GRPO and PPO, respectively, and 3.8 and 6.3

Method	Metric	2Wiki	MuSiQue	HotpotQA	Bamboogle	WebShop	ALFWorld
		@1 / @2 / @4					
Token	pass@k	24.4 / 28.6 / 31.8	13.6 / 14.6 / 24.0	19.2 / 24.0 / 24.8	24.0 / 25.6 / 28.8	**47.0** / 48.0 / 55.2	72.2 / 77.4 / **88.8**
	avg@k	24.4 / 21.4 / 24.4	13.6 / 11.6 / 11.2	19.2 / 18.3 / 17.6	24.0 / 19.8 / 22.1	**47.0** / 45.8 / 44.6	72.2 / 73.0 / 73.0
Turn	pass@k	21.2 / 33.4 / 36.0	9.8 / 12.0 / 27.4	16.2 / 18.0 / 25.4	24.8 / 32.0 / 35.2	42.2 / **56.6** / 62.6	73.6 / **81.0** / 86.0
	avg@k	21.2 / 21.5 / 24.1	9.8 / 11.7 / 9.7	16.2 / 17.3 / 17.8	24.8 / 21.1 / 25.1	42.2 / 43.4 / 44.7	73.6 / **73.7** / 72.9
GIGPO	pass@k	18.6 / 25.6 / 28.8	9.6 / 13.4 / 26.4	17.0 / 18.6 / 23.0	16.0 / 27.2 / 31.2	41.0 / 52.4 / 53.4	67.6 / 76.4 / 76.4
	avg@k	18.6 / 21.6 / 21.5	9.6 / 9.6 / 11.9	17.0 / 17.9 / 17.7	16.0 / 19.9 / 17.3	41.0 / 39.9 / 38.7	67.6 / 68.4 / 68.4
TAPO	pass@k	**25.6 / 36.8 / 37.6**	**16.0 / 16.0 / 31.0**	**21.4 / 26.2 / 28.2**	**25.6 / 36.0 / 37.6**	46.6 / 56.2 / **63.4**	**74.2** / 80.8 / 88.2
	avg@k	**25.6 / 22.8 / 25.6**	**16.0 / 14.3 / 13.6**	**21.4 / 22.4 / 22.8**	**25.6 / 27.3 / 26.5**	46.6 / **45.9 / 45.5**	**74.2** / 73.4 / **74.1**

Table 2: Per-task 4B GRPO-family recorded final reward values (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.

Method	Metric	2Wiki	MuSiQue	HotpotQA	Bamboogle	WebShop	ALFWorld
		@1 / @2 / @4					
Token	pass@k	26.6 / 33.8 / 36.0	15.2 / 17.4 / 34.4	20.4 / 26.6 / 30.0	26.4 / 33.6 / 33.6	**47.4** / 49.6 / 53.0	81.6 / 87.4 / 88.2
	avg@k	26.6 / 25.3 / 25.3	15.2 / 16.3 / 16.2	20.4 / 23.0 / 23.5	26.4 / 27.1 / 27.6	**47.4** / 44.8 / 43.3	81.6 / 81.0 / 80.2
Turn	pass@k	25.6 / 31.0 / 34.0	19.2 / 20.0 / 27.4	26.0 / 27.2 / 29.2	32.8 / 36.0 / 36.8	44.8 / **51.6** / 61.2	79.2 / 83.6 / 86.6
	avg@k	25.6 / 26.9 / 26.9	19.2 / 16.1 / 16.1	26.0 / 22.8 / 22.9	32.8 / 31.9 / 33.1	44.8 / 44.9 / 43.5	79.2 / 80.0 / **80.7**
GIGPO	pass@k	24.4 / 28.2 / 30.0	13.2 / 18.2 / 30.0	19.8 / 25.4 / 25.4	23.2 / 25.6 / 32.8	39.4 / 39.6 / 51.0	70.8 / 74.8 / 78.4
	avg@k	24.4 / 25.4 / 25.4	13.2 / 16.2 / 15.3	19.8 / 22.9 / 23.0	23.2 / 24.6 / 24.6	39.4 / 39.5 / 43.4	70.8 / 71.3 / 70.4
TAPO	pass@k	**29.0 / 35.2 / 39.6**	**21.4 / 24.2 / 37.0**	**26.6 / 29.4 / 32.0**	**36.0 / 39.2 / 44.8**	47.0 / 50.8 / **61.4**	**82.2 / 87.6 / 88.4**
	avg@k	**29.0 / 27.7 / 27.7**	**21.4 / 18.4 / 20.1**	**26.6 / 29.4 / 26.3**	**36.0 / 37.2 / 36.5**	47.0 / **45.3 / 43.6**	**82.2 / 81.3** / 80.6

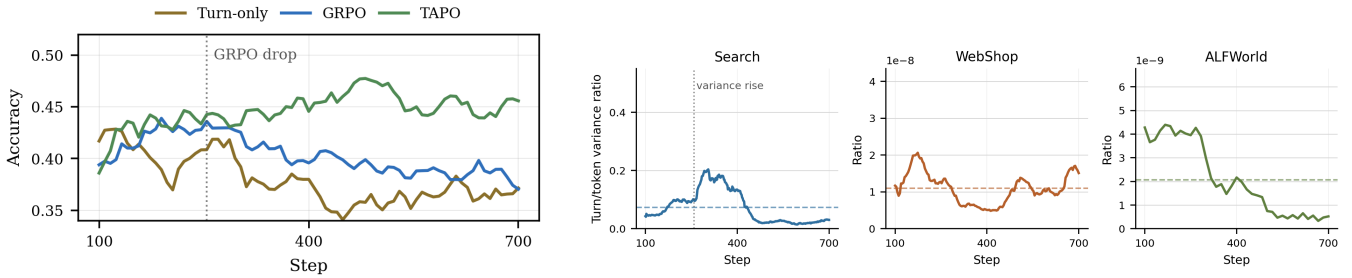
Table 3: Per-task 9B GRPO-family recorded final reward values (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.

Method	Metric	2Wiki	MuSiQue	HotpotQA	Bamboogle	WebShop	ALFWorld
		@1 / @2 / @4					
Token	pass@k	21.0 / 29.0 / 34.4	15.4 / 15.6 / 23.0	18.2 / 19.6 / 23.6	21.6 / 29.6 / 30.4	**42.6** / 53.4 / 57.2	76.0 / 76.8 / **89.2**
	avg@k	21.0 / 20.9 / **23.5**	15.4 / 15.3 / 15.3	18.2 / 17.9 / 17.9	21.6 / 20.6 / 20.7	**42.6** / 41.6 / 40.4	76.0 / 76.8 / **77.0**
Turn	pass@k	18.8 / 32.4 / 39.0	12.4 / 12.6 / 25.2	13.8 / 14.2 / 23.8	20.8 / 35.2 / 36.8	38.2 / **60.6** / 64.8	77.4 / **80.6** / 86.0
	avg@k	18.8 / 21.0 / 21.3	12.4 / 12.3 / 12.3	13.8 / 13.5 / 13.8	20.8 / 20.7 / 21.1	38.2 / 39.4 / 40.5	77.4 / 76.7 / 76.9
TAPO	pass@k	**24.8 / 39.0 / 44.6**	**21.4 / 21.6 / 31.8**	**23.6 / 23.8 / 30.0**	**26.4 / 41.6 / 43.2**	42.2 / 60.0 / **65.6**	**77.8** / 80.0 / 88.4
	avg@k	**24.8 / 26.0** / 23.4	**21.4 / 21.4 / 21.4**	**23.6 / 23.8 / 23.8**	**26.4 / 26.7 / 26.2**	42.2 / **41.8 / 40.8**	**77.8 / 76.9** / 76.8

Table 4: Per-task 4B PPO-family recorded final reward values (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.

Method	Metric	2Wiki	MuSiQue	HotpotQA	Bamboogle	WebShop	ALFWorld
		@1 / @2 / @4					
Token	pass@k	31.4 / 32.0 / 35.4	15.2 / 17.0 / 32.8	22.8 / 29.0 / **34.6**	29.6 / 32.8 / 34.4	**48.4 / 50.6** / 57.0	78.0 / **83.2 / 90.2**
	avg@k	31.4 / 27.9 / 27.4	15.2 / 17.0 / 16.6	22.8 / 25.8 / 25.8	29.6 / 28.6 / 29.9	**48.4 / 46.8** / 45.5	78.0 / **77.7** / 75.8
Turn	pass@k	28.0 / 28.0 / 31.2	20.2 / 20.6 / 25.2	**28.8** / 30.2 / 30.2	34.4 / 35.2 / 37.6	42.8 / 50.0 / 65.8	73.6 / 80.4 / 85.2
	avg@k	28.0 / 28.0 / **31.2**	20.2 / 16.9 / 17.3	**28.8** / 25.7 / 25.7	34.4 / 32.6 / 34.0	42.8 / 44.3 / **45.7**	73.6 / 75.0 / 75.9
TAPO	pass@k	**33.2 / 33.4 / 36.2**	**20.4 / 23.4 / 36.6**	26.8 / **31.0** / 34.2	**38.4 / 40.0 / 45.6**	48.0 / 49.8 / **66.6**	**78.6** / 82.4 / 89.6
	avg@k	**33.2 / 29.9** / 29.2	**20.4 / 20.4 / 21.3**	26.8 / **28.8 / 28.0**	**38.4 / 39.7 / 39.6**	48.0 / 46.7 / 45.6	**78.6** / 77.3 / **77.5**

Table 5: Per-task 9B PPO-family recorded final reward values (%). Tasks are columns and methods are rows; each task cell lists @1 / @2 / @4. Bold marks the best value for each task, metric, and k.



(a) Bamboogle checkpoints **(b)** Turn/token variance ratios

Figure 2: Training stability and variance diagnostics. (a) GRPO declines on OOD Bamboogle after the marked early-training transition, while TAPO remains stable. (b) The search variance ratio rises at the same point and has mean 7.3×10^{-2}, compared with 1.1×10^{-8} for WebShop and 2.0×10^{-9} for ALFWorld.

points for 4B. Bamboogle shows the largest model-scale averages, with TAPO gains of 9.3 points for 9B and 6.8 points for 4B, and TAPO improves the aggregate score on each in-distribution search benchmark in every model/objective configuration. On WebShop and ALFWorld, pass@1 remains within 0.6 points of Token for 9B and within 2.0 points for 4B, consistent with limited intervention in the low-variance regime.

Training Stability and Variance Diagnostics Both panels of Figure 2 use GRPO 9B training-curve data. Panel (a) shows GRPO declining on Bamboogle after the marked transition while TAPO remains stable; panel (b) shows the concurrent rise in the search turn/token variance ratio. Its mean is 7.3×10^{-2}, versus 1.1×10^{-8} for WebShop and 2.0×10^{-9} for ALFWorld. Consistently, the Turn rows in Tables 2–5 are not uniformly optimal, whereas TAPO leads every aggregate search configuration and stays close to Token at pass@1 on the low-ratio embodied tasks. These results support adapting aggregation to the observed variance regime.

6 Ablation and OOD Evaluation

6.1 Ablation Study

Tables 2–5 isolate the aggregation rule by comparing Token, fixed Turn, and adaptive TAPO. Averaged over all six tasks, both metrics, both RL families, and $k \in \{1, 2, 4\}$, TAPO improves over Token/Turn by 3.5/3.0 points for 9B and 4.0/4.1 points for 4B. Fixed turn normalization helps in some high-variance settings but is not uniformly stronger than Token; adaptation retains the useful regime of both endpoints.

6.2 Comparison with GIGPO

We use GIGPO as a practical multi-turn turn-credit baseline at the same model scales. For middle-step grouping, rollouts that reach the same turn-level state, defined by the same thinking-and-tool-use stage, form one group; we then perform GRPO-style relative grouping over these intermediate states. SeeUPO instead requires multiple backward/update passes (Hu et al. 2026), while IGPO requires additional forward likelihood evaluations (Wang et al. 2025), imposing stricter compute overhead. Across all six tasks, both metrics, and $k \in \{1, 2, 4\}$ in Tables 2 and 3, TAPO improves over GIGPO by 6.3 points at 4B and 7.9 points at 9B; the search-only gains are 6.2 and 7.4 points.

6.3 OOD Evaluation

Bamboogle contains general compositional QA questions and is excluded from the training mixture. Its broader question distribution and requirement to combine evidence make it a suitable OOD test of whether learned search and reasoning behaviors transfer beyond the training benchmarks. Averaging pass@k and avg@k for $k \in \{1, 2, 4\}$ across GRPO and PPO, TAPO scores 39.3 versus 30.1 for Token and 34.3 for Turn at 9B; at 4B, it scores 30.8 versus 24.0 and 26.6. Together with Figure 2(a), these results show improved final OOD performance and late-training stability.

7 Conclusion

Multi-turn RL with only final rewards faces a practical credit-assignment dilemma: exact turn-level credit is unavailable, while token-normalized clipped policy-gradient objectives can overweight long, highly correlated turns. We introduced TAPO, a turn-adaptive policy-optimization method that separates turn-level and token-level noise and mixes token-level and turn-level aggregation according to the estimated variance ratio. Across the latest Tables 2–5, TAPO improves over the direct Token baseline by 3.5 points for 9B and 4.0 points for 4B, with search gains of 4.9 and 5.1 points. It also improves the OOD Bamboogle aggregate from 30.1 to 39.3 for 9B and from 24.0 to 30.8 for 4B. At the same time, it remains essentially identical to GRPO in low turn-level variance-ratio settings such as WebShop and ALFWorld.

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